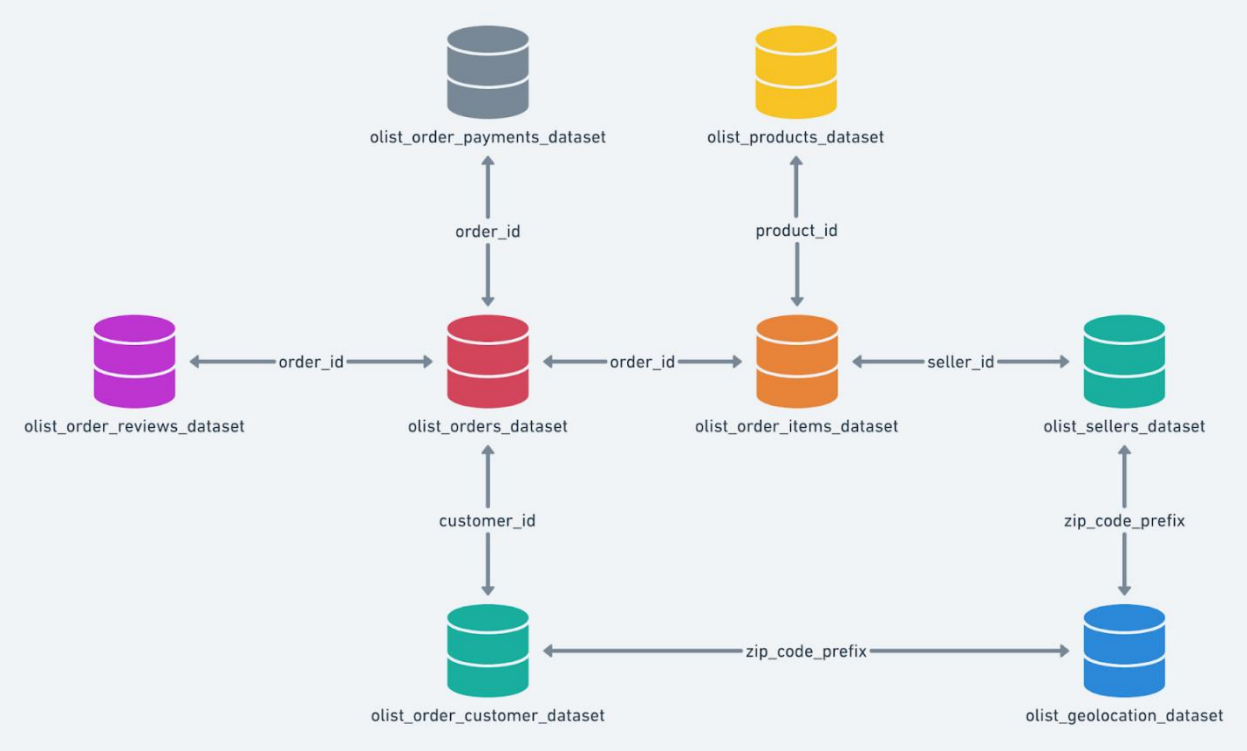
Bussines Case Study

Company Name: Target

Target is a globally renowned brand and a prominent retailer in the United States. Target makes itself a preferred shopping destination by offering outstanding value, inspiration, innovation and an exceptional guest experience that no other retailer can deliver.
This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders placed between 2016 and 2018. The dataset offers a comprehensive view of various dimensions including the order status, price, payment and freight performance, customer location, product attributes, and customer reviews.
By analyzing this extensive dataset, it becomes possible to gain valuable insights into Target's operations in Brazil. The information can shed light on various aspects of the business, such as order processing, pricing strategies, payment and shipping efficiency, customer demographics, product characteristics, and customer satisfaction levels.

Dataset Schema:



Problem Statement:

A. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1. Data type of all columns in the "customers" table.

```
Select column_name, data_type
   from scaler-dsml-sql-475411.target.INFORMATION_SCHEMA.COLUMNS
    where table_name = 'customers'
```

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

Results per page: 50 1 – 5 of 5

2. Get the time range between which the orders were placed.

```
Select min(order_purchase_timestamp) as first_order_placed,
       max(order_purchase_timestamp) as last_order_placed
      from `target.orders`
```

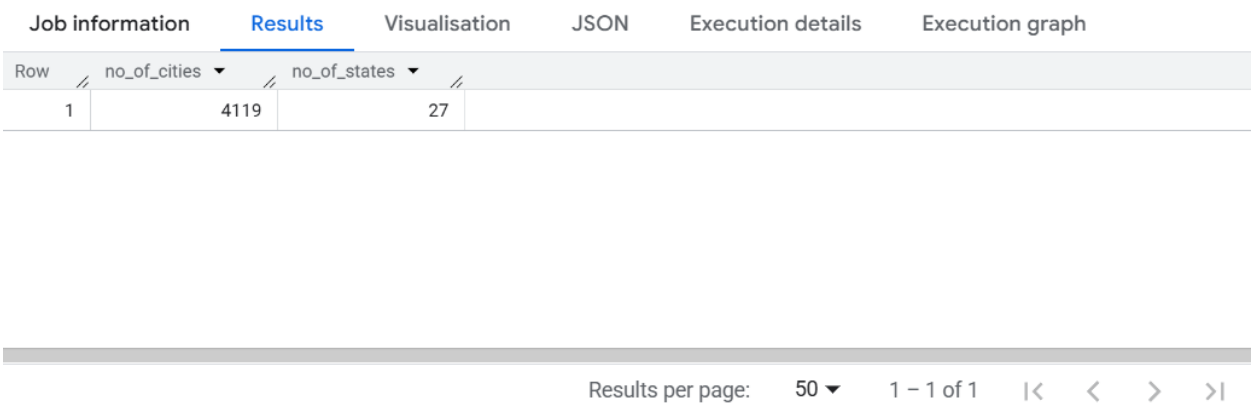
Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	first_order_placed	last_order_placed
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

Results per page: 50 1 – 1 of 1

3. Count the Cities & States of customers who ordered during the given period.

```
Select count(distinct customer_city) as no_of_cities ,
        count(distinct customer_state) as no_of_states
        from `target.customers` c
        join `target.orders` o
        on c.customer_id = o.customer_id
```



Insight:- This illustrates Target's market presence in urban and rural areas of Brazil.

B. In-depth Exploration:

1. Is there a growing trend in the no. of orders placed over the past years?

```
Select FORMAT_DATE('%Y-%m', order_purchase_timestamp) as
        Year_Month, count(*) as no_of_orders
         from `target.orders`
        group by 1
         order by 1
```

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Year_Month	no_of_orders
1	2016-09	4
2	2016-10	324
3	2016-12	1
4	2017-01	800
5	2017-02	1780
6	2017-03	2682
7	2017-04	2404
8	2017-05	3700
9	2017-06	3245
10	2017-07	4026

Results per page: 50 1 – 25 of 25

Graphical Representation:



Insight:- Shows whether Target's Brazilian operations are growing, stable, or declining year over year.

2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

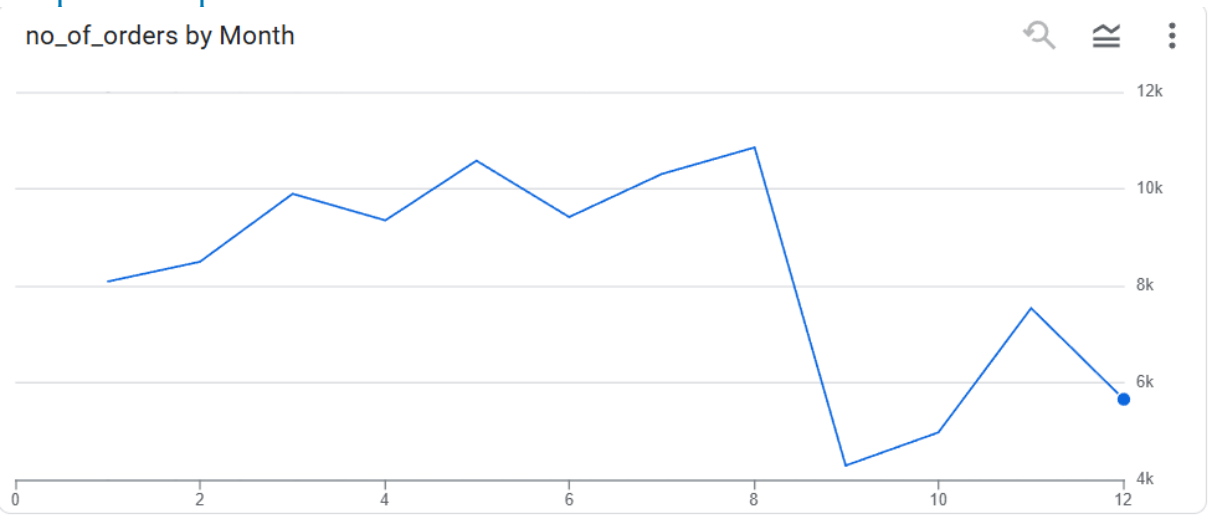
```
Select extract(month from order_purchase_timestamp) as Month,
       count(*) as no_of_orders
       from `target.orders`
       group by 1
       order by 1
```

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Month	no_of_orders
1	1	8069
2	2	8508
3	3	9893
4	4	9343
5	5	10573
6	6	9412
7	7	10318
8	8	10843
9	9	4305
10	10	4959

Results per page: 50 | 1 – 12 of 12

Graphical Representation:



Insight:- Shopping activity gradually increases from January to August, reaching its peak during this period. After August, the no. of orders suddenly declines.

3. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)
    1. 0-6 hrs : Dawn
    2. 7-12 hrs : Mornings
    3. 13-18 hrs : Afternoon
    4. 19-23 hrs : Night

```sql
select case when extract(hour from order_purchase_timestamp)
            between 0 and 6 then 'Dawn'

            when extract(hour from order_purchase_timestamp)
            between 7 and 12 then 'Morning'

            when extract(hour from order_purchase_timestamp)
            between 13 and 18 then 'Afternoon'

            when extract(hour from order_purchase_timestamp)
            between 19 and 23 then 'Night'

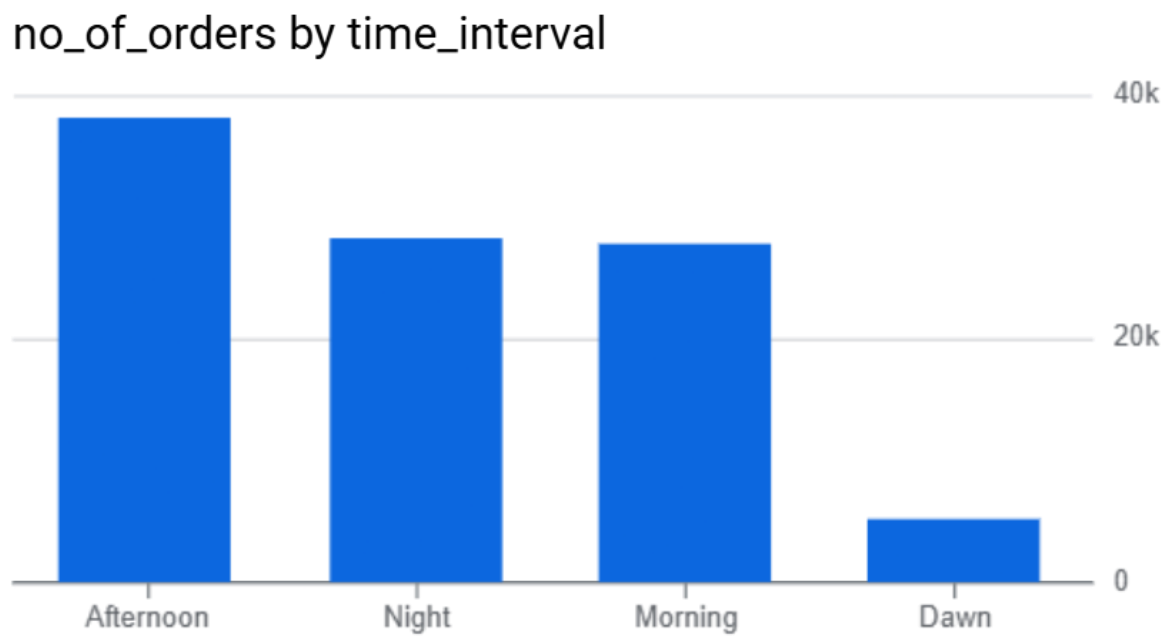
            end as time_interval,
            count(order_id) as no_of_orders
            from `target.orders`
            group by 1
            order by 2 desc
```

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	time_interval	no_of_orders
1	Afternoon	38135
2	Night	28331
3	Morning	27733
4	Dawn	5242

Results per page: 50 1 – 4 of 4

Graphical Representation:



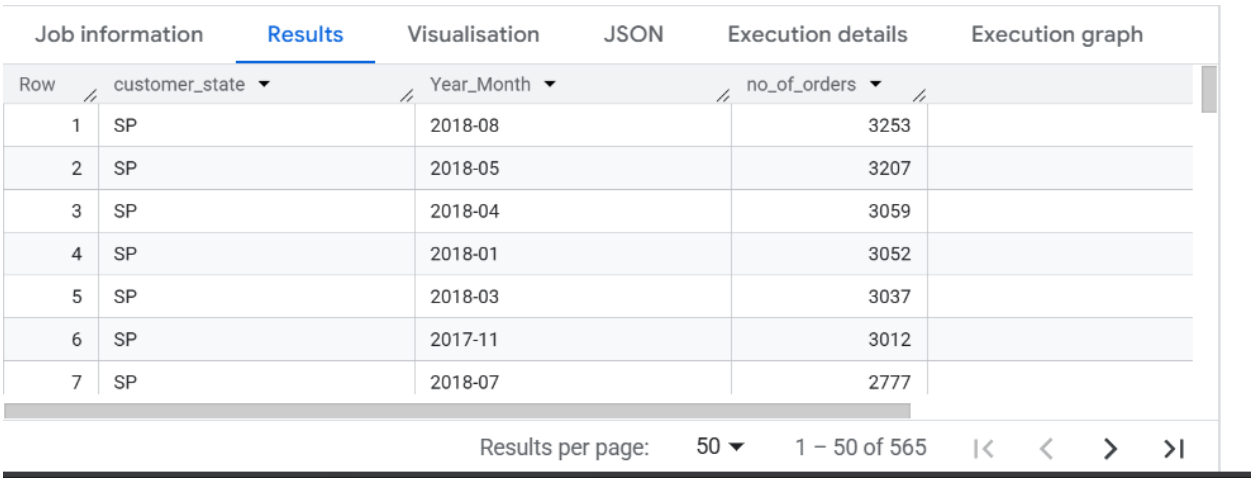
Insight:- Knowing when Brazilian customers prefer to shop online is important for planning marketing campaigns and managing server load.

C. Evolution of E-commerce orders in the Brazil region:

1. Get the month on month no. of orders placed in each state.

```
Select c.customer_state,
        format_date ('%Y-%m',order_purchase_timestamp)
            as Year_Month,
            count(o.order_id) as no_of_orders
            from `target.customers` c
            join `target.orders` o
            on o.customer_id = c.customer_id
            group by 1,2
            order by 3 desc
```

Graphical Representation:



Job information | Results | Visualisation | JSON | Execution details | Execution graph

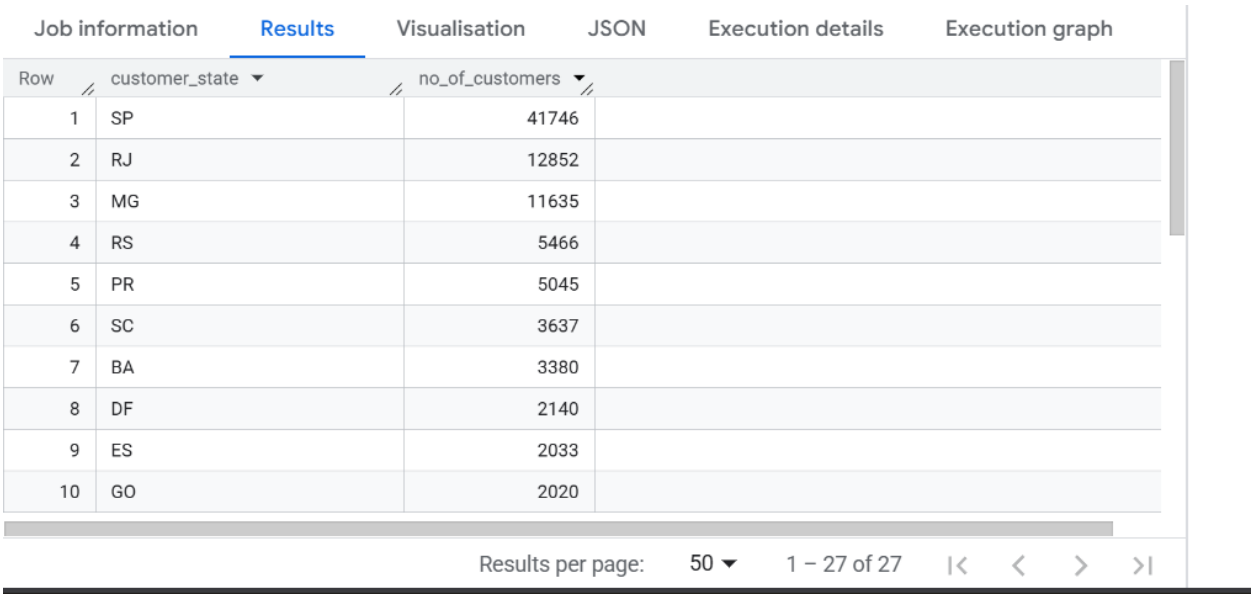
Row	customer_state	Year_Month	no_of_orders
1	SP	2018-08	3253
2	SP	2018-05	3207
3	SP	2018-04	3059
4	SP	2018-01	3052
5	SP	2018-03	3037
6	SP	2017-11	3012
7	SP	2018-07	2777

Results per page: 50 1 – 50 of 565

Insight:- The number of orders varies by state each month, showing which states have the most active online shoppers.

2. How are the customers distributed across all the states?

```
Select c.customer_state,
       count(distinct c.customer_id) as no_of_customers
       from `target.customers` c
       join `target.orders` o
       on o.customer_id = c.customer_id
       group by 1
       order by 2 desc
```



Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	customer_state	no_of_customers
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020

Results per page: 50 | 1 – 27 of 27

Insight:- Most customers are concentrated in a few states, while other states have fewer online shoppers.

D. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

```
with final as (
Select extract (year from o.order_purchase_timestamp) as year,
       sum (p.payment_value) as total_cost
       from `target.orders` o
       join `target.payments` p
       on o.order_id = p.order_id
       where extract (year from o.order_purchase_timestamp) in
       (2017,2018) and
       Extract (month from o.order_purchase_timestamp) between
       1 and 8
        group by 1)

           Select year, total_cost,
           ((lead(total_cost) over (order by year) -
           total_cost)/total_cost)*100 as percentage_yoy
```

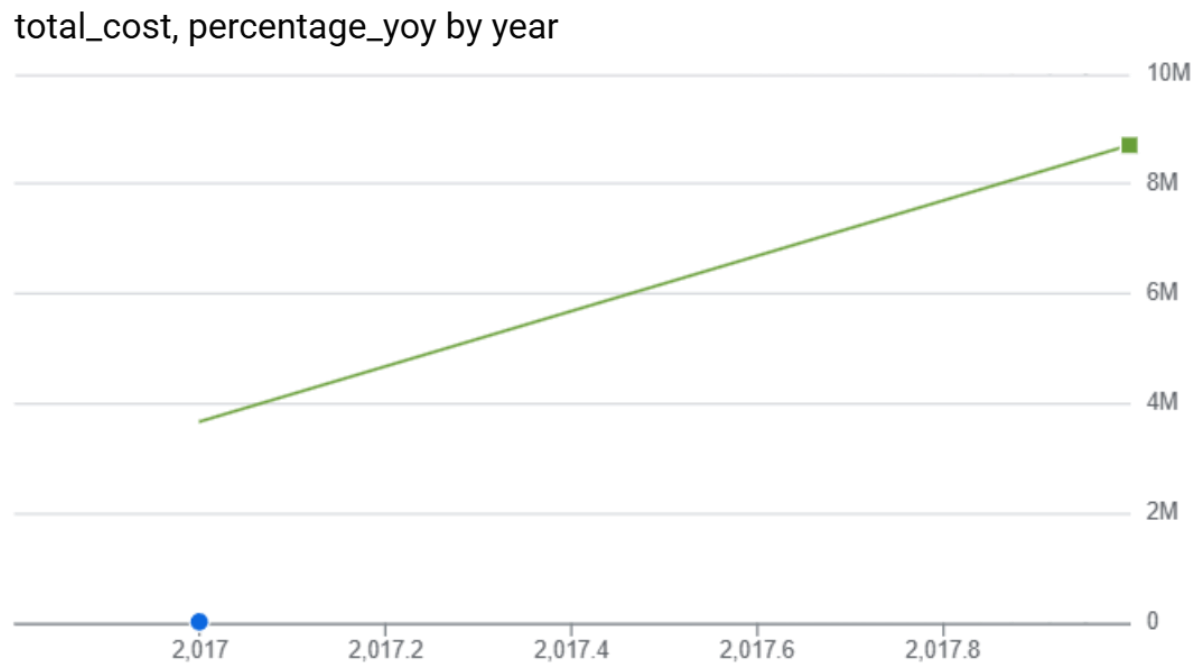
```
from final
order by 1
```

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	year	total_cost	percentage_yoy
1	2017	3669022.119999...	136.9768716466...
2	2018	8694733.840000...	null

Results per page: 50 | 1 – 2 of 2

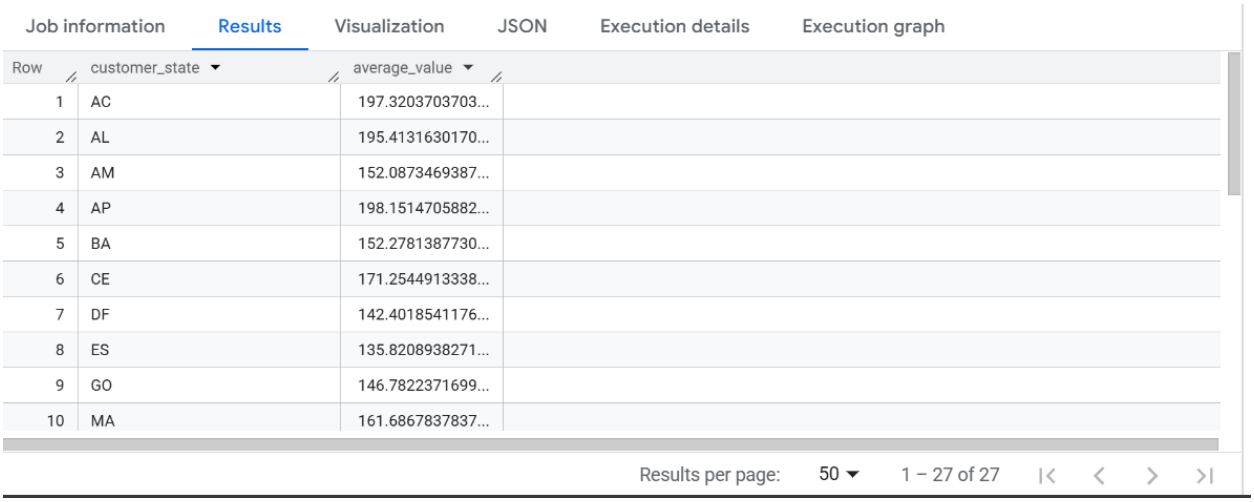
Graphical Representation:



Insight:- Spending from January to August increased from 2017 to 2018, showing stronger economic activity driven by online shopping.

2. Calculate the Total & Average value of order price for each state.

```
Select c.customer_state, (sum(price)/count(distinct order_id))
       as average_value
      from `target.customers` c
      join `target.orders` o
      using(customer_id)
      join `target.order_items`
      using (order_id)
      group by 1
       order by 2 desc
```

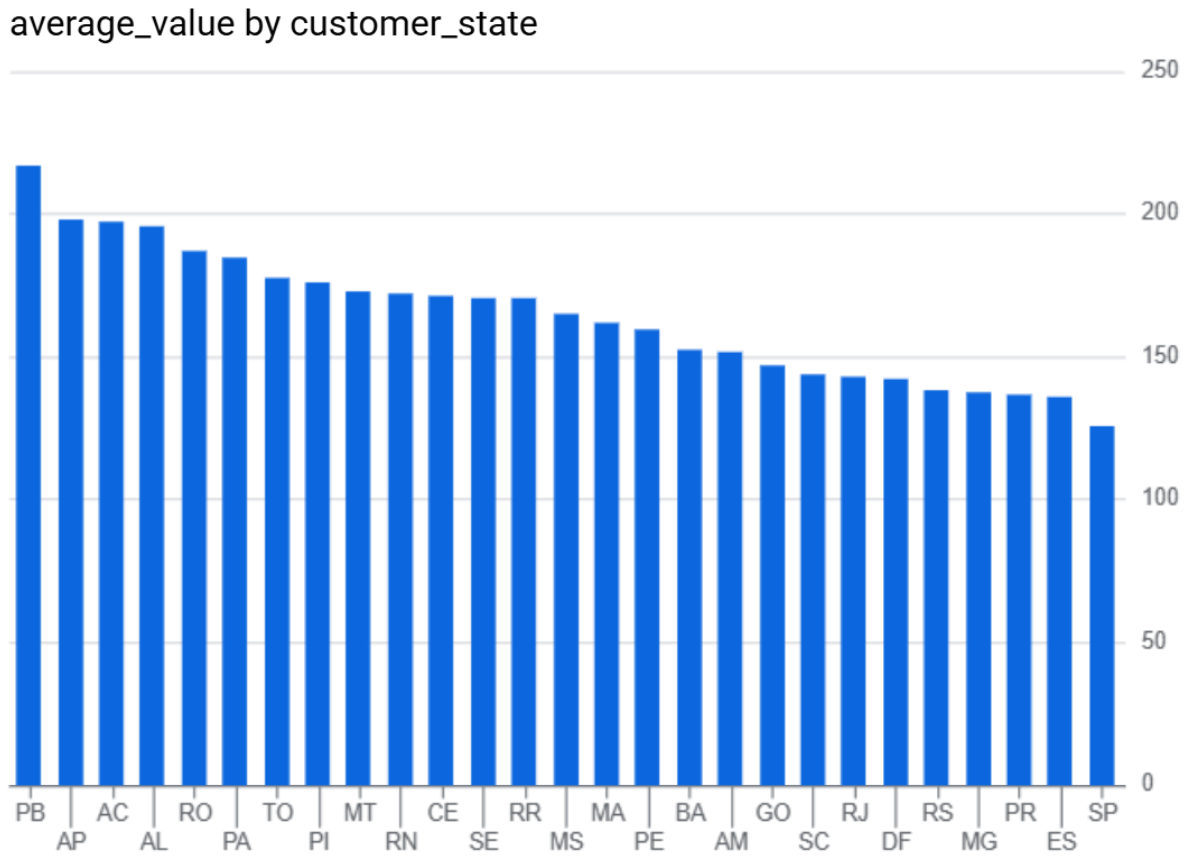


Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	customer_state	average_value
1	AC	197.3203703703...
2	AL	195.4131630170...
3	AM	152.0873469387...
4	AP	198.1514705882...
5	BA	152.2781387730...
6	CE	171.2544913338...
7	DF	142.4018541176...
8	ES	135.8208938271...
9	GO	146.7822371699...
10	MA	161.6867837837...

Results per page: 50 1 – 27 of 27

Graphical Representation:



3. Calculate the Total & Average value of order freight for each state.

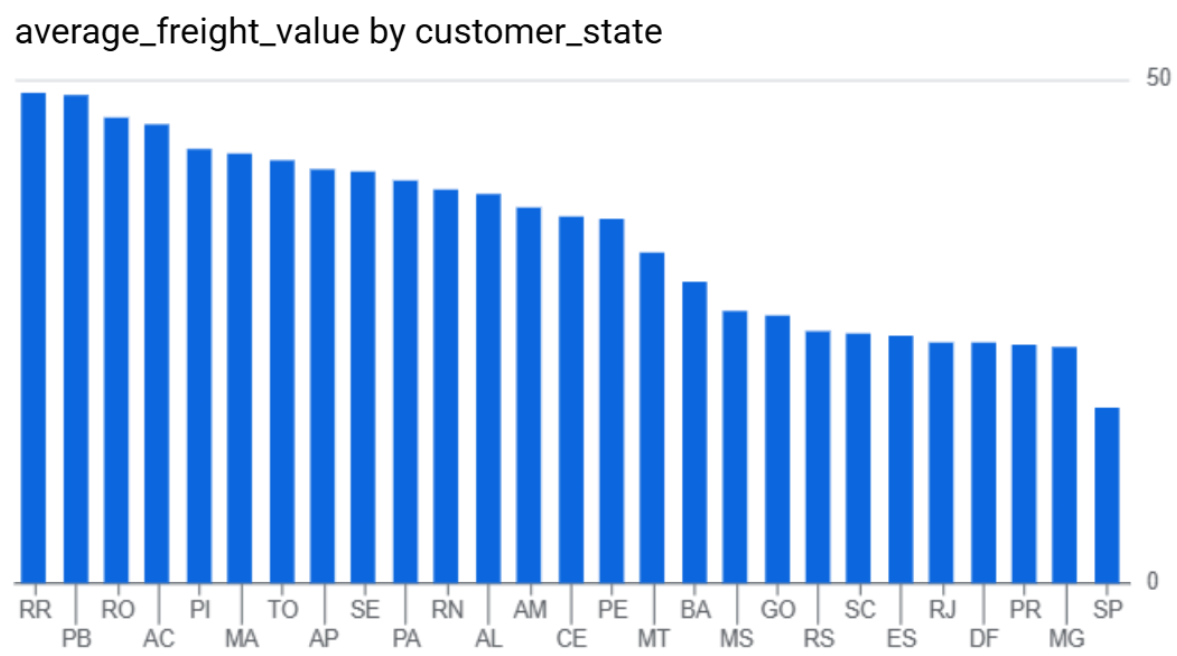
```
Select c.customer_state, (sum(freight_value)/count(distinct
       order_id)) as average_freight_value
      from `target.customers` c
      join `target.orders` o
      using(customer_id)
      join `target.order_items`
      using (order_id)
      group by 1
      order by 2 desc
```

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	customer_state	average_freight_v...
4	AC	45.51543209876...
5	PI	43.03894523326...
6	MA	42.59968918918...
7	TO	42.05261648745...
8	AP	41.00735294117...
9	SE	40.90281159420...
10	PA	39.89618556701...
11	RN	39.12883817427...

Results per page: 50 1 – 27 of 27

Graphical Representation:



E. Analysis based on sales, freight and delivery time.

1.Find the no. of days taken to deliver each order from the order's purchase date as delivery time.
Also, calculate the difference (in days) between the estimated & actual delivery date of an order.
Do this in a single query.

```
Select order_id,
    date_diff (order_delivered_customer_date,
    order_purchase_timestamp, day) as time_to_deliver,

    date_diff(order_estimated_delivery_date,
    order_delivered_customer_date, day) as
    diff_estimated_delivery

    from `target.orders`
    where order_delivered_customer_date is not null


    order by 2 desc
```

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	order_id	time_to_deliver	diff_estimated_d...
1	ca07593549f1816d26a572e06...	209	-181
2	1b3190b2dfa9d789e1f14c05b6...	208	-188
3	440d0d17af552815d15a9e41a...	195	-165
4	285ab9426d6982034523a855f...	194	-166
5	0f4519c5f1c541ddec9f21b3bd...	194	-161
6	2fb597c2f772eca01b1f5c561bf...	194	-155
7	47b40429ed8cce3aee9199792...	191	-175
8	2fe324febf907e3ea3f2aa96508...	189	-167
9	2d7561026d542c8dbd8f0daea...	188	-159
10	c27815f7e3dd0b926b5855262...	187	-162

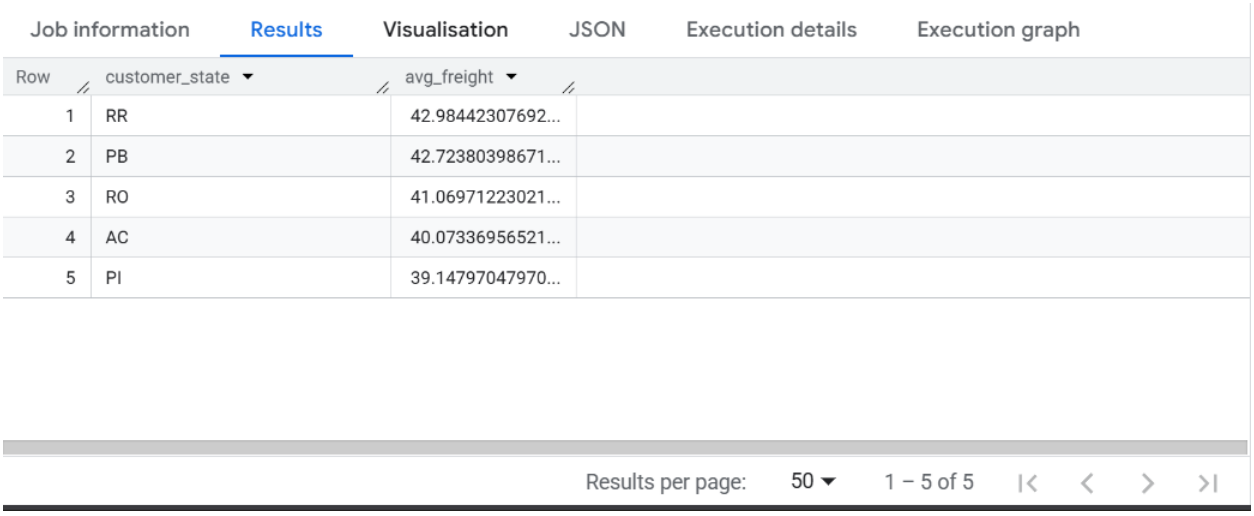
Results per page: 50 | 1 – 50 of 96476

Insight:- Orders usually take a few days to reach customers, but in some cases, delivery can take over a month.

2.Find out the top 5 states with the highest & lowest average freight value.

Top 5 Highest average Freight value.

```
Select customer_state, avg(freight_value) as
       avg_freight
       from `target.customers` c
       join `target.orders` o
       using(customer_id)
     join `target.order_items` oi
     using (order_id)
     group by 1
     order by 2 desc
     limit 5
```



Job information | Results | Visualisation | JSON | Execution details | Execution graph

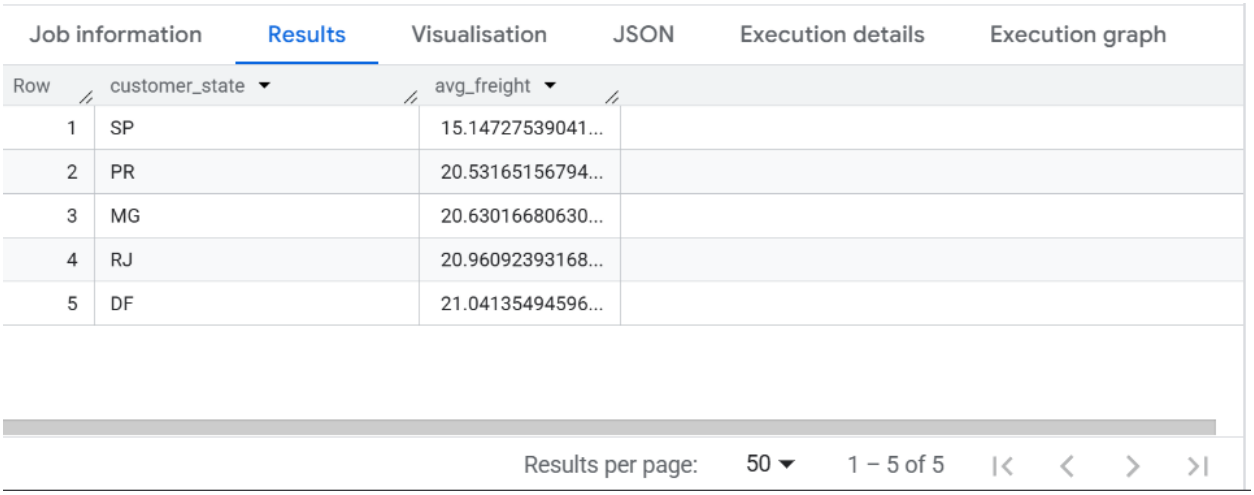
Row	customer_state	avg_freight
1	RR	42.98442307692...
2	PB	42.72380398671...
3	RO	41.06971223021...
4	AC	40.07336956521...
5	PI	39.14797047970...

Results per page: 50 | 1 – 5 of 5

Top 5 Lowest average Freight value.

```
Select customer_state, avg(freight_value) as
               avg_freight
               from `target.customers` c
               join `target.orders` o
               using(customer_id)
               join `target.order_items` oi
               using (order_id)
               group by 1
```

```
order by 2
limit 5
```



Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	customer_state	avg_freight
1	SP	15.14727539041...
2	PR	20.53165156794...
3	MG	20.63016680630...
4	RJ	20.96092393168...
5	DF	21.04135494596...

Results per page: 50 | 1 – 5 of 5

3.Find out the top 5 states with the highest & lowest average delivery time.

Top 5 Highest average delivery time

```
with final as
  (Select customer_state,
       date_diff (order_delivered_customer_date,
       order_purchase_timestamp, day) as delivery_days,
       from `target.customers` c
       join `target.orders` o
       using(customer_id)
       where order_delivered_customer_date is not null)


          select customer_state, avg(delivery_days) as
                 avg_delivery_days
                 from final
                 group by 1
                 order by 2 desc
                 limit 5
```

Job information | Results | Visualisation | JSON | Execution details | Execution graph

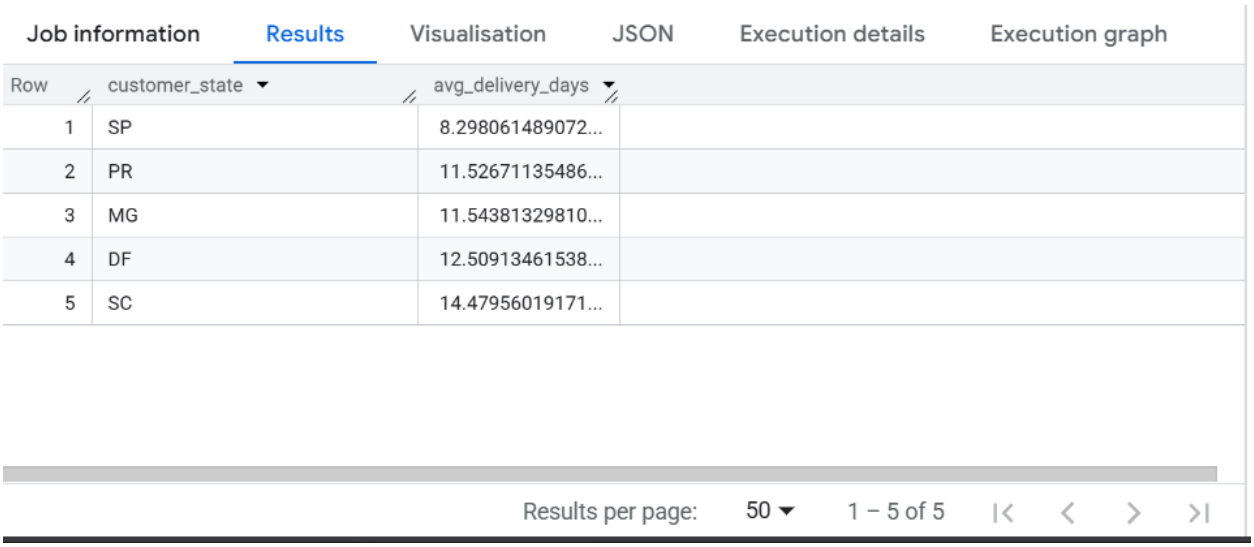
Row	customer_state	avg_delivery_days
1	RR	28.97560975609...
2	AP	26.73134328358...
3	AM	25.98620689655...
4	AL	24.04030226700...
5	PA	23.31606765327...

Results per page: 50 1 – 5 of 5

Top 5 Lowest average delivery time

```
with final as
  (Select customer_state,
       date_diff (order_delivered_customer_date,
       order_purchase_timestamp, day) as delivery_days,
       from `target.customers` c
       join `target.orders` o
       using(customer_id)
       where order_delivered_customer_date is not null)


          select customer_state, avg(delivery_days) as
                  avg_delivery_days
                  from final
                  group by 1
                  order by 2
                  limit 5
```



Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	customer_state	avg_delivery_days
1	SP	8.298061489072...
2	PR	11.52671135486...
3	MG	11.54381329810...
4	DF	12.50913461538...
5	SC	14.47956019171...

Results per page: 50 | 1 – 5 of 5

4.Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

```
with final as
(Select c.customer_state,
     date_diff(order_estimated_delivery_date,
     order_delivered_customer_date, day) as
       diff_estimated_delivery
       from `target.customers` c
       join `target.orders` o
       using(customer_id)
       where  order_delivered_customer_date is not null)

       select customer_state, avg(diff_estimated_delivery) as
              average_days
              from final
              group by 1
              order by 2 desc
              limit 5
```

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	customer_state	average_days
1	AC	19.7625
2	RO	19.13168724279...
3	AP	18.73134328358...
4	AM	18.60689655172...
5	RR	16.41463414634...

Results per page: 50 1 – 5 of 5

F. Analysis based on the payments:

1. Find the month on month no. of orders placed using different payment types.

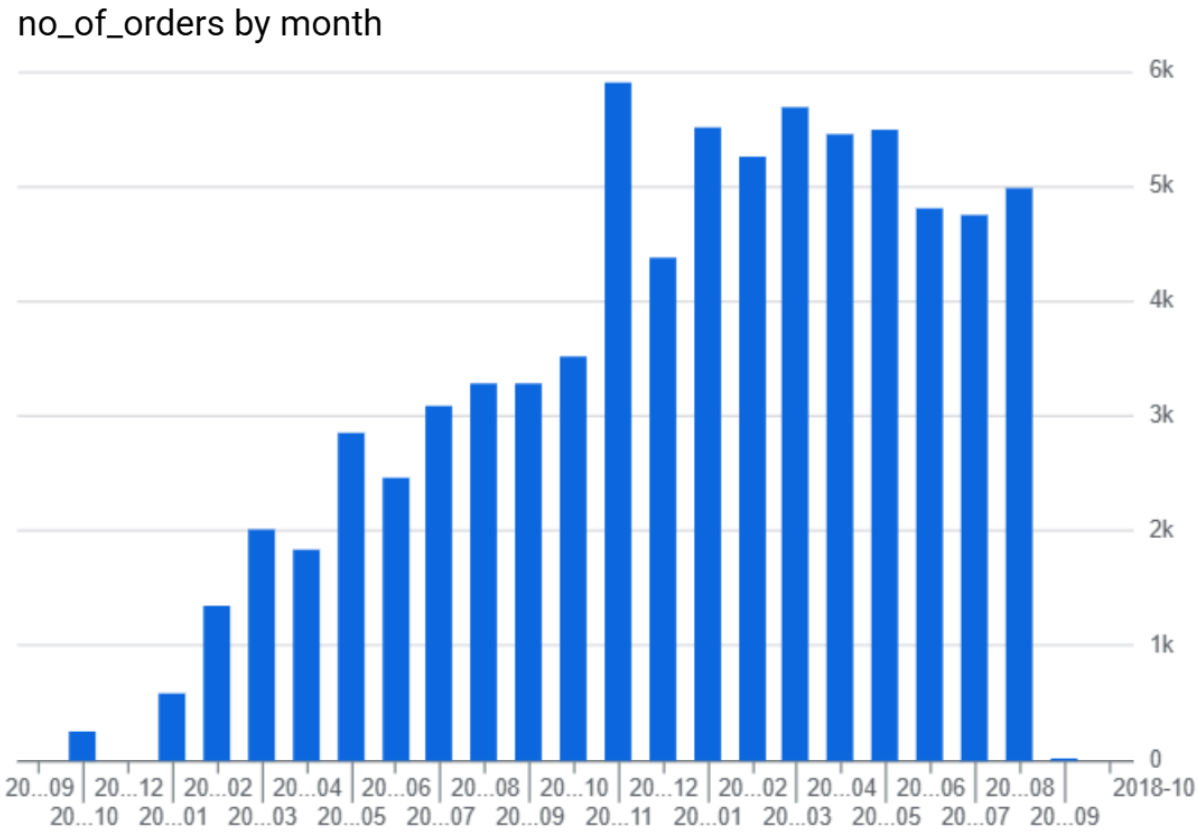
```
Select FORMAT_DATE('%Y-%m', order_purchase_timestamp)
        as month, count(*) as no_of_orders, payment_type
        from `target.orders` o
        join `target.payments` p
        using(order_id)
        group by 1,3
        order by 1
```

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	month	no_of_orders	payment_type
1	2016-09	3	credit_card
2	2016-10	63	UPI
3	2016-10	2	debit_card
4	2016-10	23	voucher
5	2016-10	254	credit_card
6	2016-12	1	credit_card
7	2017-01	583	credit_card
8	2017-01	61	voucher
9	2017-01	9	debit_card
10	2017-01	197	UPI

Results per page: 50 1 – 50 of 90

Graphical Representation:



Insight:- Credit cards are the most used payment method every single month.

2.Find the no. of orders placed on the basis of the payment installments that have been paid.

```
Select
    count(order_id) as no_of_orders,
          payment_installments
          from `target.payments`
           where payment_installments >= 1
           group by 2
           order by 1
```

Row	no_of_orders	payment_installments
1	1	22
2	1	23
3	3	21
4	5	16
5	8	17
6	15	14
7	16	13
8	17	20
9	18	24
10	23	11

Results per page: 50 1 – 23 of 23

Insight:- Most orders are paid in a few installments, while fewer orders use many installments.

Key Insights

- Orders in Brazil increased every year, with strong growth seen from 2017 to 2018.
- Most orders are placed in the afternoon and evening, shows that customers typically shop after work or during their free time.
- Customers in remote and less-connected states pay more for shipping and wait longer for deliveries. This shows there are logistics challenges in these areas.
- Credit cards are the most popular way to pay, and many customers like to use installment plans. This shows how important it is to offer flexible payment options.
- In many states, customers receive their orders before the estimated delivery date. Whereas in some states, delivery takes more than a month. Which shows that company needs to identify the reasons behind for these delays.

Actionable Recommendations

- Since most customers shop in the evening and at night, company should promote special offers and discounts during these hours.
- Target should open warehouses closer to states where shipping costs are high so that products can be delivered faster and at a lower cost. It will also help to deliver orders faster.
- Delivery timelines should be improved by using past delivery data so customers get more accurate delivery dates.
- Easy EMI and installment options should be clearly shown to encourage customers to buy higher-priced items.
- The company should also bring in more sellers and products in low order states to reach new customers and to capture the uncovered market.
- The company should promote its fast and on-time deliveries in advertisements to build trust among customers.
